

CFB TRENTON, ON, Canada

WMO: 716210

Lat: 44.1178N

Lon: 77.5238W

Elev: 283

StdP: 14.55

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.5	-1.9	-18.7	2.0	-6.1	-12.6	2.8	-0.5	29.0	26.0	25.9	25.7	7.0	340	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.3	84.8	72.0	82.2	70.6	79.9	69.3	74.5	81.5	72.9	79.5	71.3	77.5	10.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	120.4	79.0	70.6	113.8	77.3	69.1	108.0	75.7	38.2	81.5	36.7	79.5	35.3	77.7	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	23.2	20.2	17.7	DB	-15.3	89.3	6.7	3.5	-20.1	91.8	-24.0	93.9	-27.8	95.8	-32.6	98.4
(5)				WB	-15.9	77.4	6.5	2.0	-20.6	78.8	-24.4	80.0	-28.1	81.1	-32.8	82.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	46.0	19.9	22.1	30.9	43.6	55.5	65.2	69.7	68.3	61.2	49.3	37.8	27.4	(6)
(7)		DBStd	19.37	12.58	10.62	9.80	7.90	7.27	6.14	5.00	5.04	6.91	7.62	8.39	10.27	(7)
(8)		HDD50	3742	933	781	595	220	27	0	0	0	4	108	373	701	(8)
(9)		HDD65	7341	1398	1201	1059	643	308	72	12	24	154	488	816	1166	(9)
(10)		CDD50	2293	0	0	1	27	197	456	612	569	339	86	6	0	(10)
(11)		CDD65	419	0	0	0	0	13	78	159	128	39	2	0	0	(11)
(12)		CDH74	2745	0	0	0	5	92	502	1068	832	241	5	0	0	(12)
(13)		CDH80	494	0	0	0	0	10	94	211	142	37	0	0	0	(13)
(14)	Wind	WSAvg	7.8	9.1	8.9	8.7	8.9	7.7	7.0	6.9	6.3	6.5	7.5	8.2	8.6	(14)
(15)	Precipitation	PrecAvg	35.7	3.0	2.4	2.4	3.0	3.1	3.0	3.2	2.7	3.3	3.0	3.3		(15)
(16)		PrecMax	44.5	5.0	4.9	4.5	5.7	5.0	6.5	9.0	4.5	7.6	8.1	5.2	5.5	(16)
(17)		PrecMin	27.3	1.8	1.0	0.9	0.4	0.9	0.2	0.6	0.9	0.9	1.1	0.5	1.1	(17)
(18)		PrecStd	4.5	0.9	0.9	1.1	1.3	1.1	1.7	1.6	1.1	1.4	1.5	1.2	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	48.0	60.2	72.9	80.9	86.6	88.1	87.5	84.1	73.8	61.6	54.1	(19)
(20)		MCWB	47.9	43.5	50.8	60.1	67.7	72.2	73.8	72.2	72.5	65.1	55.6	50.8		(20)
(21)		2%	DB	44.2	43.3	52.8	65.9	76.5	82.7	85.0	83.7	80.2	69.3	57.4	47.9	(21)
(22)		MCWB	42.1	39.9	45.5	53.5	64.6	69.7	72.4	71.6	71.0	62.9	53.2	45.4		(22)
(23)		5%	DB	40.0	40.0	48.3	60.6	72.6	79.6	82.2	80.9	76.4	65.6	54.4	44.5	(23)
(24)		MCWB	38.0	37.2	42.7	50.2	61.8	68.4	70.8	70.1	68.3	59.8	50.3	41.9		(24)
(25)		10%	DB	36.7	37.1	44.3	56.6	68.4	76.6	79.6	78.6	73.4	62.0	50.8	40.9	(25)
(26)		MCWB	34.5	34.4	39.5	47.8	58.9	66.6	69.1	69.2	66.1	57.1	47.0	38.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.5	44.2	53.2	61.3	70.2	74.7	77.0	76.6	74.6	67.8	57.1	51.3	(27)
(28)		MCDWB	49.8	46.2	58.4	70.0	78.1	83.0	85.0	84.1	81.7	71.5	59.7	53.1		(28)
(29)		2%	WB	42.2	40.5	47.0	56.0	66.4	72.3	74.6	74.2	72.0	63.7	54.2	46.0	(29)
(30)		MCDWB	43.7	43.1	51.1	63.3	74.0	79.9	81.9	80.3	78.0	67.9	57.0	47.6		(30)
(31)		5%	WB	38.2	37.5	43.2	52.0	63.2	70.2	72.8	72.6	69.7	60.6	50.8	42.2	(31)
(32)		MCDWB	39.8	39.6	47.3	56.7	70.2	77.1	79.6	78.4	75.1	64.6	53.6	44.3		(32)
(33)		10%	WB	34.7	34.8	39.8	49.1	60.5	68.1	71.0	70.9	67.3	57.7	47.3	38.5	(33)
(34)		MCDWB	36.4	37.0	44.0	55.2	67.2	74.4	77.2	76.6	72.1	61.5	50.5	40.7		(34)
(35)	Mean Daily Temperature Range	MDBR	15.6	16.1	15.8	17.5	18.4	17.5	17.3	17.8	18.6	16.1	14.3	13.2		(35)
(36)		5% DB	MCDBR	15.2	16.3	20.8	24.8	22.9	20.6	19.4	20.2	20.8	19.8	19.3	15.3	(36)
(37)		MCWBR	14.6	14.6	15.1	15.4	13.2	11.3	10.6	11.2	12.7	14.1	16.3	14.0	(37)	
(38)		5% WB	MCDBR	15.3	15.6	19.5	21.7	20.5	18.1	16.4	16.6	18.6	18.4	17.6	14.9	(38)
(39)		MCWBR	15.0	14.6	15.3	15.8	13.2	11.2	10.2	10.3	12.0	14.3	15.9	14.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.296	0.306	0.321	0.358	0.381	0.422	0.421	0.410	0.371	0.352	0.311	0.286	(40)	
(41)		taud	2.398	2.362	2.429	2.386	2.371	2.280	2.294	2.314	2.411	2.453	2.517	2.485	(41)	
(42)		Ebn at Noon	263	280	292	287	282	270	269	267	270	261	257	258	(42)	
(43)		Edh at Noon	24	30	32	36	38	42	41	38	32	27	22	20	(43)	
(44)	All-Sky Solar Radiation	RadAvg	421	689	1112	1429	1735	1849	1885	1651	1322	782	484	335	(44)	
(45)		RadStd	43	65	101	156	172	148	147	89	109	69	58	43	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page